Evaluating Information Retrieval Systems

CSC 618: Information Retrieval, Week 10, Self Study 2, Part A
Utica University, Fall 2026

1. Why retrieval systems need evaluation

Evaluation helps measure and compare the effectiveness of different retrieval systems.

Without it, we can only guess from a few searches whether a change, such as adding stemming or moving from TF-IDF to BM25, really helps. One change can improve some queries and hurt others, so systems are scored on the same queries and the same judgments to keep the comparison fair.

2. Relevance judgments and ground-truth datasets

Before evaluating the search engine, we need to know which docs are considered relevant to a particular query. This is called relevance judgement.

- Ground truth datasets contain expected or manually judged correct dataset which is used to evaluate the search engine.
- Relevance is judged against the user's information need, so a doc can contain every query word and still not be relevant.
- Large collections cannot be judged in full. TREC judges only a pool of top results from several systems, and unjudged docs are usually counted as non-relevant.

3. Precision, Recall and F1-score

Precision = TP / (TP + FP); how many retrieved docs are actually relevant.

Recall = TP / (TP + FN); how many of all available relevant documents the search engine successfully retrieves.

F1-score = 2 × (Precision × Recall) / (Precision + Recall); measure that balances precision & recall using harmonic mean.

These three ignore ranking order. Precision and recall usually trade off, since returning more docs tends to raise recall and lower precision. The harmonic mean stays low if either one is low, so a system cannot get a high F1 by doing well on only one.

4. Precision@K and Reciprocal Rank

Precision@K: how many of the top k retrieved docs are relevant.

Reciprocal Rank (RR): measures how high the first relevant doc. appears in the search results.

P@K ignores the order inside the top k and everything below it. RR looks only at the first relevant doc, so it suits tasks where one correct answer is enough. Averaging RR over many queries gives Mean Reciprocal Rank (MRR).

5. Average Precision and Mean Average Precision

- Average Precision (AP): calculates precision at every position where a relevant doc. appears, then averages those values over all relevant documents. (One query)
- Mean Average Precision (MAP): calculates AP across multiple search queries.
- A relevant doc that is never retrieved counts as 0, so AP drops when docs are missed or ranked low. MAP gives every query the same weight.

6. Graded relevance and nDCG

Grade the relevance rather than treating relevance as relevant (1) and irrelevant (0).

nDCG uses these grades in three steps. Each doc's grade is its gain. The gain is divided by log2(rank + 1), so a good doc placed low earns less, and adding these values gives DCG. DCG is then divided by the DCG of the ideal ranking, which keeps the score between 0 and 1 and makes queries comparable.

7. Why evaluation metrics can lead to different conclusions

Different metrics are used for measuring different aspects of search engine performance. Only one aspect of measure does not provide full picture. One score can be perfect & we measure using another aspect then it can perform badly.

For example, take a query with 3 relevant docs. System 1 places them at ranks 1, 7 and 8 (RR = 1.00, AP = 0.55). System 2 places them at ranks 2, 3 and 4 (RR = 0.50, AP = 0.64). System 1 wins on RR and System 2 wins on AP, so the choice of metric should follow the task.

Metric definitions and formulas

Symbol / metric	Definition
TP	True positives: relevant documents that were retrieved
FP	False positives: retrieved documents that are not relevant
FN	False negatives: relevant documents that were not retrieved

Precision	TP / (TP + FP)
Recall	TP / (TP + FN)
F1	2 × Precision × Recall / (Precision + Recall)
P@K	(number of relevant documents in the top K results) / K
RR	1 / (rank of the first relevant document); 0 if none is retrieved
MRR	(1 / |Q|) × Σ RR(q) over all queries q in Q
AP	(1 / |Rel|) × Σ P@k over every rank k that holds a relevant document
MAP	(1 / |Q|) × Σ AP(q) over all queries q in Q
DCG@K	Σ rel_i / log2(i + 1) for i = 1 to K, where rel_i is the relevance grade at rank i
IDCG@K	DCG@K of the ideal ranking (documents sorted by grade, highest first)
nDCG@K	DCG@K / IDCG@K, between 0 and 1

|Q| = number of queries; |Rel| = number of relevant documents for the query; Σ = sum.

Worked example

The query is MatchAllDocsQuery, which returns every document, and the information need is “chicken dishes”. It returns 17 results, and 5 of them are relevant, at ranks 5, 6, 8, 10 and 12. So TP = 5 (relevant recipes retrieved), FP = 12 (retrieved recipes that are not relevant) and FN = 0 (no relevant recipes missed).

Metric	Formula filled in	Result
Precision	TP / (TP + FP) = 5 / (5 + 12) = 5/17	0.294
Recall	TP / (TP + FN) = 5 / (5 + 0) = 5/5	1.000
F1	2 × 0.294 × 1.0 / (0.294 + 1.0) = 0.588 / 1.294	0.455
P@3	relevant in top 3 ÷ 3 = 0/3	0.000
RR	1 ÷ rank of first relevant = 1/5	0.200
AP	(1/5 + 2/6 + 3/8 + 4/10 + 5/12) / 5 = (0.200 + 0.333 + 0.375 + 0.400 + 0.417) / 5 = 1.725 / 5	0.345

The F1 result uses the exact precision of 5/17. The rounded values shown in its row would give 0.454. These values match the Self Study 1 metrics table (0.294, 1.0, 0.455, 0, 0.345).

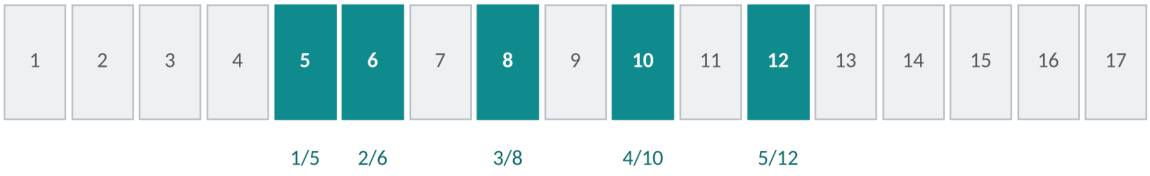


Figure 1. The 17 ranked results. Teal boxes are the relevant recipes, with the precision at that rank below each one.

AP works like this: each time the list reaches a relevant recipe, ask what fraction of everything so far is relevant. At rank 8, we have seen 3 chicken recipes out of 8 results, so 3/8. AP is the average of those five fractions.

Perfect recall is not impressive here, because a query that returns everything finds all the relevant docs without doing any real search. Precision of 0.294 means that about 7 in 10 recipes a user scrolls past are not chicken dishes. P@3 = 0 and RR = 0.2 mean the first four results show no chicken dish at all, and the first one only appears at rank 5. TermQuery on the same need scored 1.0 on every metric, so the two queries tie on recall while the other metrics separate them. This is my topic 7 point: judged by recall alone MatchAll looks perfect, but the other metrics show it is a poor result for the user.

Summary of study resources

Stanford IR book: Evaluation in information retrieval

This chapter explains how to measure whether a search system works. It describes what a test collection contains, then covers measures for unranked results (precision, recall, F-measure) and ranked results (MAP, P@K, NDCG). It also explains how judgments are collected through pooling and checked with the kappa statistic, and reminds us that user satisfaction depends on more than relevance.

Video: Evaluation measures for search and recommender systems

James Briggs explains Recall@K, MRR, MAP@K and NDCG@K using eight images and the query “cat in a box”, then builds each metric in Python. He separates offline metrics from online metrics such as A/B tests, and says offline testing comes first. He also shows that Recall@K ignores order and can be made perfect by raising K, while NDCG needs graded labels.

NIST TREC datasets

TREC, run by NIST, provides test collections made of documents, topics and relevance judgments, along with the trec_eval scoring tool and data from tracks such as Web, Deep Learning, Clinical Trials and RAG. Benchmark collections matter because everyone tests on the same data with the same scoring, so results from different systems and research groups can be compared.

Conclusion

Search quality cannot be judged reliably from a few example queries or from how a system feels to its developer. Systematic evaluation fixes the documents, queries and judgments, so a change in score can be traced to the system itself. It also makes us decide what a good result means for the task, since the MatchAll example scores perfectly on recall and poorly on ranking. Offline scores should still be checked with real users, because they say nothing about speed or satisfaction.